UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

PART A (10 MARKS)

Answer ALL questions.

1. What are parallel arrays?
   A. Arrays with no index
   B. Arrays whose elements at the same index are related
   C. Arrays that hold only characters
   D. Arrays stored in different programs

2. Which loop is guaranteed to execute its body at least once?
   A. for
   B. if
   C. while
   D. do...while

3. Which declaration creates a floating-point variable named price?
   A. bool price;
   B. char price;
   C. float price;
   D. int price;

4. Which statement immediately exits a loop?
   A. continue
   B. return 0
   C. break
   D. cin

5. Which term describes a finite sequence of precise steps that solves a problem?
   A. Compiler
   B. Variable
   C. Algorithm
   D. Array

6. Which data type stores a whole number?
   A. string
   B. float
   C. char
   D. int

7. Which operator sends data to cout?
   A. ==
   B. =
   C. <<
   D. >>

8. What does a compiler do?
   A. Executes one source statement at a time
   B. Stores only integer values
   C. Translates the whole source program before execution
   D. Draws flowcharts automatically

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9. What does the % operator return?
   A. The square root
   B. The absolute value
   C. The remainder of integer division
   D. The quotient

10. Which part of a for loop is executed after each iteration?
   A. The initialisation
   B. The loop body twice
   C. The condition only
   D. The update expression

PART B (60 MARKS)

QUESTION 1

This question tests programming concepts and error terminology.

a) Define compiler and interpreter.

(2 marks)

b) State THREE differences between a compiler and an interpreter.

(3 marks)

QUESTION 2

A program reads a positive integer N and calculates the sum and average of the numbers 1 to N.

a) Draw a flowchart for the sum and average calculation.

(6 marks)

b) Write pseudocode and a complete C++ program for the algorithm.

(7 marks)

QUESTION 3

This question tests operator precedence and assignment.

a) Evaluate the following expressions: (i) 7 + 3 * 2, (ii) (7 + 3) * 2, (iii) 17 % 5, (iv) 17 / 5, (v) 17.0 / 5.

(5 marks)

b) Write C++ statements to declare two integers a and b, read their values, and display their sum, difference, product and remainder.

(6 marks)

QUESTION 4

Study the following array program segment:

```
int a[5] = {2, 4, 6, 8, 10};
int s = 0;
for (int i = 0; i < 5; i++) {
   s += a[i];
}
cout << s;
```

a) Trace the program and state the output.

(6 marks)

b) Rewrite the segment using a do...while loop that sums only values greater than 5.

(9 marks)

QUESTION 5

A sports club stores sport names and medal counts in parallel arrays.

a) Declare two parallel arrays for five sports and initialise them with suitable values.

(6 marks)

b) Write a program segment to display each sport with its medals, then calculate the total and average medals.

(10 marks)

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PART C (30 MARKS)

QUESTION 1

A lecturer records the marks of ten students in an array.

Write a complete C++ program that reads ten marks, displays the average, the highest mark and its index, and counts how many marks are above the average.

(15 marks)

QUESTION 2

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

END OF QUESTION PAPER